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- **Problem Statement ID –** SIH26098
- **Problem Statement Title –** Development of a Low-Cost Precision Guidance and Smart Electronic Fuze System for a 155 mm Artillery Shell
- **Theme –** Smart Vehicles
- **PS Category –** Hardware
- **Team ID – 149807**
- **Team Name – VEL**



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Proposed Solution

- Rotating nose fuze with fixed canards, modelled for 7 degrees of freedom.
- Alternator based actuation for accurate manoeuvring of the projectile with meticulously designed control loop to keep in account the challenges with a spin stabilized projectile.
- Kalman filter enabled sensor architecture to get accurate data against spaces between data rates, cuts cost as well as helps with accuracy in harsh conditions.
- Microcontroller programmed to calculate ideal trajectory and correcting the projectile to get to the correct path using sensor enabled control loops.
- All the mentioned details in the problem statement have been taken into consideration and all the required files are here: https://drive.google.com/drive/folders/1aKDZRcnG_TksS9KXy8e6u6BXwWTamvus?usp=sharing



Fig 1: Reference 155mm Artillery, source: munitionsindia.in

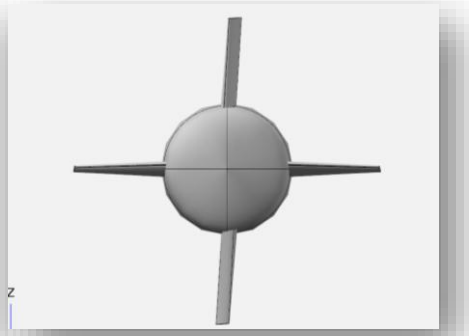


Fig 2: Front View of PGK, displaying the rotating and lift fin pairs.

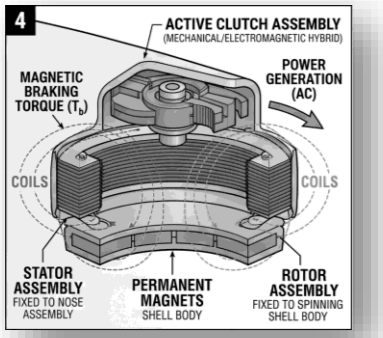


Fig 3: Representation of Alternator that generates energy and is responsible for braking.

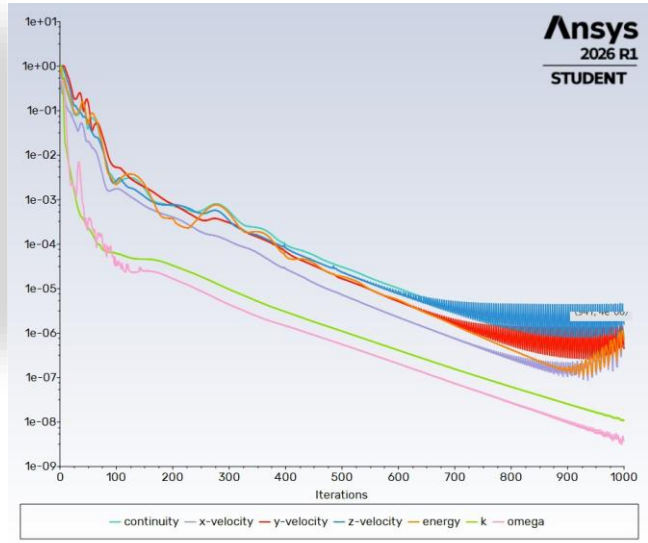


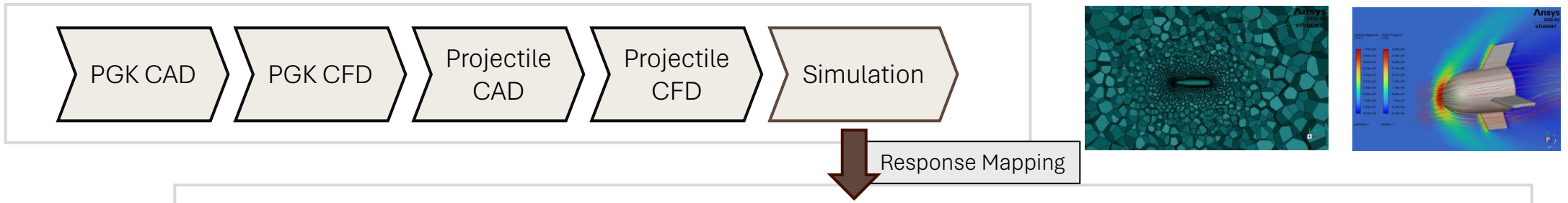
Fig 4: CFD plots of the coefficients

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TECHNICAL APPROACH

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Mathematical Modelling



Control System

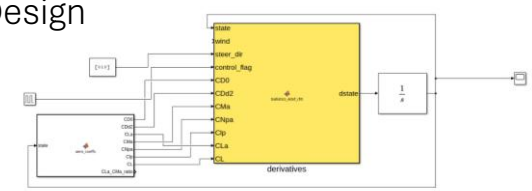
Drivelink:

https://drive.google.com/drive/folders/1aKDZRcnG_TksS9KXy8e6u6BXwWTamvus?usp=sharing

Youtube:

https://youtu.be/xyvWUJGeEc8

Alternator Design



Tools used for simulation are OpenVSP, OpenFoam, C++, ROS2-Gazebo, Python Sim, ngspice, Lepton-Schematic, GNU Octave, MATLAB-Simulink, AutoCad, ANSYS, Solidworks.

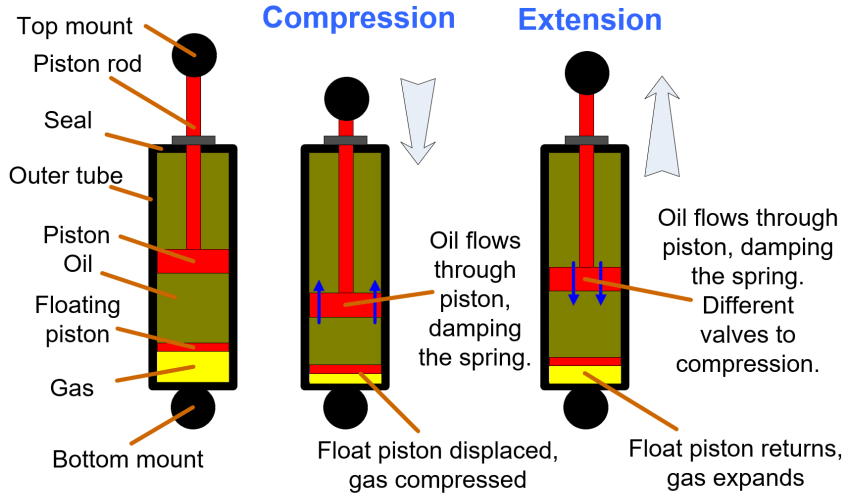
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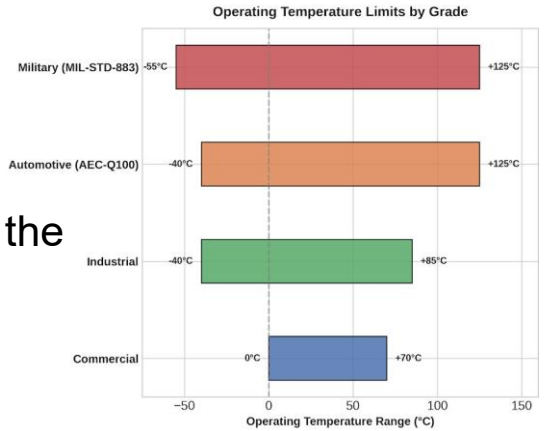
FEASIBILITY AND VIABILITY

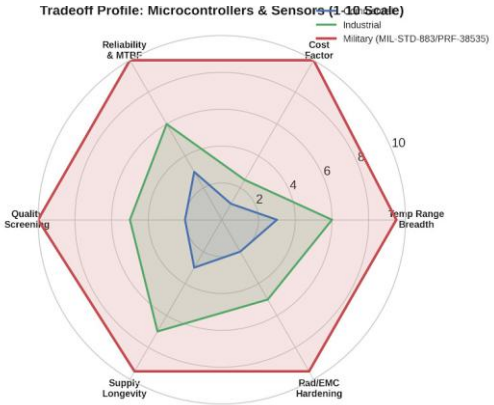
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Feasibility:
- Made by all industrial grade sensors, brings down cost about 5-10x as comepared to using military grade sensors. Precision and lower data rates are countered by Kalamn filter infusion. Robustness completely resting upon the encasing investment (Epoxy shock damping encasing).
- The achieved design works on non-rapid transient model, thus cannot destabalize the fired artillery mid air.

Potential challenges and risks and their solutions:
Challenge: The overall CEP depends a lot on the launch to account yaw of repose and corillis.
Counter: Having the launcher to already predict the estimated drift.
Challenge: GPS jammers used on the projectile.
Counter: Whole trajectory judgement shifts to IMU which considers the Kalman Coefficient built along the way to maintain the trajectory on correct path.







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IMPACT AND BENEFITS

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Impact on users (artillery units)

- Enables safer, more confident fire support closer to friendly forces or in urban/complex terrain.
- Delivers reliable first-round effects with fewer shells, reducing exposure to counter-battery fire and logistics needs.
- Upgrades existing ammo stocks for point-target precision without major system changes.



Key benefits

- **Sharply cuts collateral damage** by concentrating effects on the intended target.
- **Cost-effective**: Much cheaper than purpose-built guided rounds; leverages current inventory.
- Improves accuracy, reduces rounds needed per mission, and adds reliability (including failsafe features and anti-jam upgrades in newer versions).

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RESEARCH AND REFERENCES

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Modern Exterior Ballistics The Launch And Flight Dynamics Of Symmetric Projectiles

- https://archive.org/details/ModernExteriorBallisticsTheLaunchAndFlightDynamicsOfSymmetricProjectiles2ndEd.R.McCoy

Flight Stability of Canard-Guided Dual-Spin Projectiles with Angular Rate Loops

- https://onlinelibrary.wiley.com/doi/10.1155/2020/2705175

Efficiency Analysis of Canards-Based Course Correction Fuze for a 155-mm Spin-Stabilized Projectile

- https://ascelibrary.org/doi/10.1061/%28ASCE%29AS.1943-5525.0000634

Novel Aiming Method for Spin-Stabilized Projectiles with a Course Correction Fuze Actuated by Fixed Canards

- https://www.mdpi.com/2079-9292/8/10/1135

On Guidance and Control for Guided Artillery Projectiles, Part 1: General Considerations

- https://www.foi.se/rest-api/report/FOI-R--3291--SE

Artillery-Ammunition

- https://munitionsindia.in/wp-content/uploads/2-Artillery-Ammunition.pdf

AN-761 Electronic Fuzing

- https://www.ti.com/lit/an/snoa217/snoa217.pdf?ts=1790532646645&ref_url=https%253A%252F%252Fwww.google.com%252F

Recent Advancements in Proximity Fuzes Technology

- https://www.ijert.org/recent-advancements-in-proximity-fuzes-technology

Ballistics Theory and Design of Guns and Ammunition

- https://ftp.idu.ac.id/wpcontent/uploads/ebook/tdg/MILITARY%20PLATFORM%20DESIGN/Ballistics%20%20Theory%20and%20Design%20of%20Guns%20and%20Ammunition,%20Third%20Edition.pdf

Systems to measure yaw, spin and muzzle velocity of projectiles, improve fire control fidelity, and reduce shot-to-shot dispersion in both conventional and airbursting programmable projectiles

- https://patents.google.com/patent/US9879963B2/en

Aerodynamic and angular motion characteristics of a fixed-canard dual-spin projectile under different forebody spinning rates

- https://www.sciencedirect.com/science/article/pii/S1000936125005941?ref=pdf_download&fr=RR-2&rr=a434ab3579e6ff70